L25 Induced system norm:

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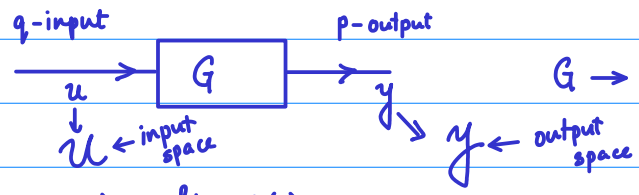
$G : \mathcal{U} \xrightarrow{\text{Banach space}} \mathcal{Y}$

$\|G\|$ ← norm



$G \to p \times q$ matrix.

Further, $G(s)$ is strictly proper, i.e. $\lim_{s\to\infty} G(s) = 0$.

$$y(t) = g(t) * u(t) = \int_0^t g(\tau)\, u(t-\tau)\, d\tau$$

$g(t) \to$ convolutional kernel

$$g(t) = \begin{bmatrix} g_{11}(t) & g_{12}(t) & \cdots & g_{1q}(t) \\ g_{21}(t) & g_{22}(t) & \cdots & g_{2q}(t) \\ \vdots & \vdots & \ddots & \vdots \\ g_{p1}(t) & g_{p2}(t) & \cdots & g_{pq}(t) \end{bmatrix} =: \begin{bmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_q(t) \end{bmatrix}$$

$$Y(s) = \underset{\substack{\uparrow \\ \text{Transfer matrix}}}{G(s)}\, U(s)$$

Input $u(t)$	Output $y(t)$	Signal norms	Induced norms
$\mathcal{L}_2$	$\mathcal{L}_2$	$\|u\|_2$	$\|G\|_\infty$
S	S	$\|u\|_S$	$\|G\|_\infty$
S	P	$\|u\|_P$	$\|G\|_2$
P	P	$\|u\|_P$	$\|G\|_\infty$
$\mathcal{L}_\infty$	$\mathcal{L}_\infty$	$\|u\|_\infty = \sup_t \max_i \lvert u_i(t)\rvert$ $\|u\|_\infty = \sup_t \|u(t)\|$	$\max_i \|g_i\|_1$ $\leq \int_0^\infty \|g(t)\|\, dt$

Computation of $\mathcal{L}_2$ and $\mathcal{H}_2$ norms:

$$\|G\|_2 = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{Trace}\left\{G^*(j\omega)\, G(j\omega)\right\} d\omega}$$

$$= \sqrt{\int_{-\infty}^{\infty} \mathrm{Trace}\left\{g^*(t)\, g(t)\right\} dt}$$

$\mathcal{L}_2(-\infty,\infty) \cong \mathcal{L}_2(j\mathbb{R})$

$g(t) \longleftrightarrow G(s)$

$\boxed{G(s) \in RH_2}$ $\qquad \mathcal{L}_2[0,\infty) \cong H_2 \supseteq RH_2$

Consider a transfer matrix $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right]$ with A stable. Then we have

$$\|G\|_2^2 = \text{trace}\left(B^* W_o B\right) = \text{trace}\left(C W_c C^*\right)$$

where W_o & W_c are the observability Gramian & controllability Gramian which can be obtained by solving

$$AW_c + W_c A^* + BB^* = 0 \checkmark, \qquad A^* W_o + W_o A + C^* C = 0. \checkmark$$ Lyapunov equation

Proof: $\because$ G is stable, $g(t) = \begin{cases} Ce^{At}B, & t \geq 0 \\ 0, & t < 0 \end{cases}$

$$\|G\|_2^2 = \int_0^\infty \text{trace}\left\{g(t)\, g^*(t)\right\} dt = \int_0^\infty \text{trace}\left\{C e^{At} BB^* e^{A^* t} C^*\right\} dt$$

Define $W_c := \int_0^\infty e^{At} BB^* e^{A^* t}\, dt \longleftarrow$ Controllability Gramian

$$AW_c + W_c A^* = A \int_0^\infty e^{At} BB^* e^{A^* t} dt + \left(\int_0^\infty e^{At} BB^* e^{A^* t} dt\right) A^*$$

$$= \int_0^\infty \left(Ae^{At} BB^* e^{A^* t} + e^{At} BB^* e^{A^* t} A^*\right) dt = \int_0^\infty \frac{d}{dt}\left(e^{At} BB^* e^{A^* t}\right) dt$$

$$= e^{At} BB^* e^{A^* t}\Big|_0^\infty = 0 - BB^*$$

$$\therefore \; AW_c + W_c A^* + BB^* = 0$$

$$\|G\|_2^2 = \text{trace}\left(C W_c C^*\right), \text{ where } AW_c + W_c A^* + BB^* = 0$$

$H_2 \leftarrow RH_2$ $\qquad \boxed{\mathcal{L}_2}$ $\qquad G \in R\mathcal{L}_2$ but $G \notin RH_2$ $\qquad \mathcal{L}_2 = H_2 \oplus H_2^\perp$

$$G(s) = [G(s)]_{RH_2} + [G(s)]_{RH_2^\perp} \qquad \boxed{\|G(s)\|_2^2 = \|[G(s)]_{RH_2}\|^2 + \|[G(s)]_{RH_2^\perp}\|^2}$$

$G(s) \in \mathcal{RL}_2$ $\qquad G(s) = G_1(s) + G_2(s)$ $\qquad G_1(s) \in \mathcal{RH}_2$, $G_2(s) \in \mathcal{RH}_2^{\perp}$

$$\|G(s)\|_2^2 = \underbrace{\|G_1(s)\|_2^2}_{\mathcal{RH}_2} + \underbrace{\|G_2(s)\|_2^2}_{\mathcal{RH}_2^{\perp}}$$

$$\underbrace{\|G_2(s)\|_2^2}_{} = \underbrace{\|G_2(-s)\|_2^2}_{\mathcal{RH}_2}$$

$$= \|G_1(s)\|_2^2 + \|G_2(-s)\|_2^2$$

e.g. $$G(s) = \begin{bmatrix} 1/s+2 & 0 \\ 0 & 1/(s+2)(s-1) \end{bmatrix} = \underbrace{\overset{G_1(s)}{\begin{bmatrix} 1/s+2 & 0 \\ 0 & 2/s+2 \end{bmatrix}}}_{\mathcal{RH}_2} + \underbrace{\overset{G_2(s)}{\begin{bmatrix} 0 & 0 \\ 0 & 1/s-1 \end{bmatrix}}}_{\mathcal{RH}_2^{\perp}}$$

$(A_1, B_1, C_1, 0)$

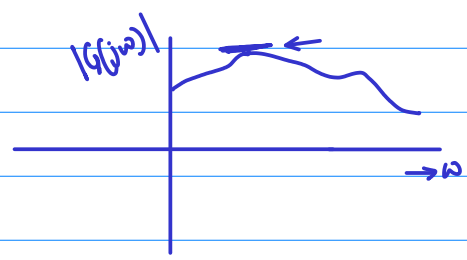
$$G_2(-s) = \overset{G_2(-s)}{\begin{bmatrix} 0 & 0 \\ 0 & \frac{-1}{s+1} \end{bmatrix}} \in \mathcal{RH}_2$$

$(A_2, B_2, C_2, 0)$

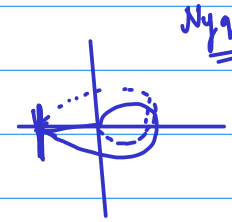
$\mathcal{L}_\infty$ and $\mathcal{H}_\infty$ norms:

$G(s) \in \mathcal{L}_\infty$ $\qquad \|G\|_\infty = \operatorname{ess\,sup}_{\omega} \sigma_{max}\{G(j\omega)\}$

$G(s) \rightarrow$ scalar transfer function



Nyquist



$\omega_1, \omega_2 \ldots\ldots \omega_N \leftarrow$ grid of frequency

$$\max_{1 \le k \le N} \sigma_{max}\{G(j\omega_k)\}$$

Let $\gamma > 0$ and

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \in \mathcal{RL}_\infty \quad \text{and} \quad \sigma_{max}(D) < \gamma$$

Then $\|G\|_\infty < \gamma$ iff H has no imaginary axis eigenvalues, where

$$H := \begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix} \quad \text{and} \quad R := \gamma^2 I - D^*D$$

Proof: